

2016.0 RANGE ROVER (LG), 501-30

REAR END SHEET METAL REPAIRS

REAR WHEELHOUSE OUTER

[G 1 4 5 4 6 4 6]

REMOVAL AND INSTALLATION

REMOVAL

NOTES:

- The rear wheelhouse outer is installed in conjunction with:
 - Quarter panel
 - Rear lamp panel outer
 - Right hand indicated, left hand similar.
-

1. 



E143600

The rear wheelhouse outer is serviced as indicated.

-
2. Before commencing this procedure make sure that you are aware of all Health and Safety requirements.

For additional information, refer to: [Health and Safety Precautions](#) (100-00 General Information, Description and Operation).

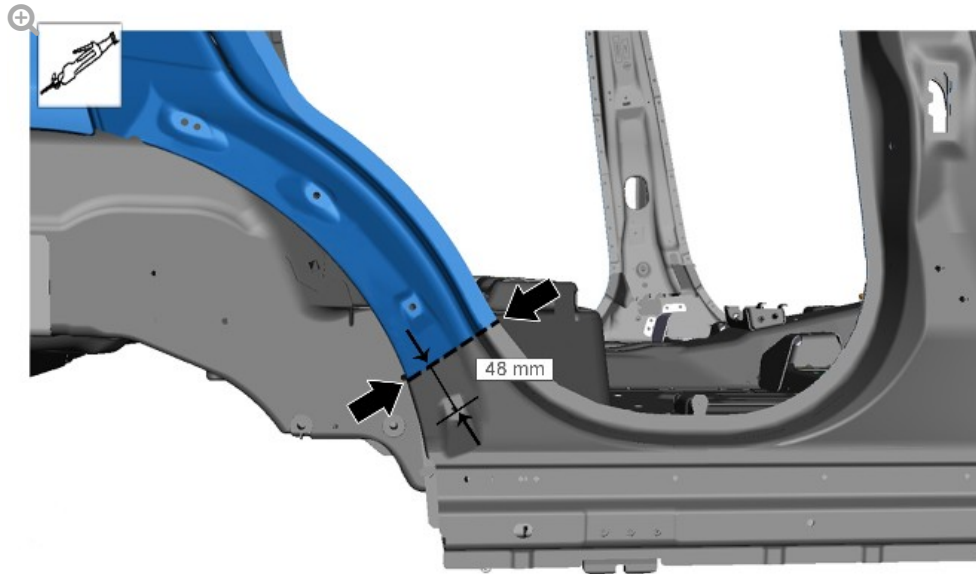
-
3. For further information on the methods, tools and fixings used in this procedure refer to the body repairs - general information section.

NOTES:

- Where breakstem fasteners (BSF) removal is instructed remove fastener shaft with a 4mm punch and remove the head with a 6.5mm drill bit. Make sure all the drilled holes are free from debris.
- Where self piercing rivets (SPR) removal is instructed use either the approved SPR tool or 5.3mm drill bit.
- Any joints sealed as part of vehicle manufacture must be sealed as part of the repair.

4. For additional information, refer to: [Quarter Panel](#) (501-30 Rear End Sheet Metal Repairs, Removal and Installation).
5. For additional information, refer to: [Rear Lamp Panel Outer](#) (501-30 Rear End Sheet Metal Repairs, Removal and Installation).
6. Release the rear wheelhouse outer wiring harness and position it to one side.

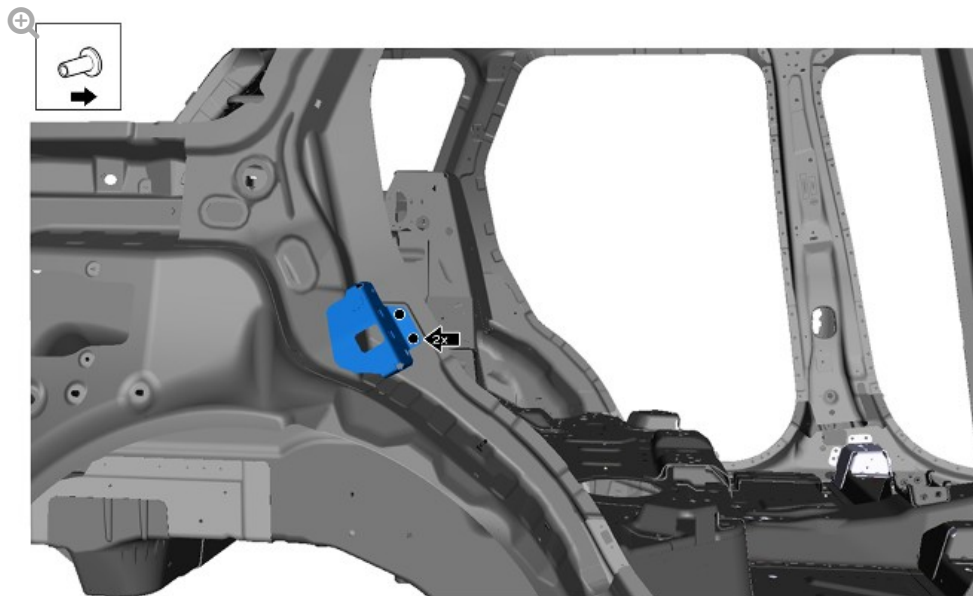
7.



E 143599

Cut the quarter panel as indicated to allow access to the SPR's.

8.



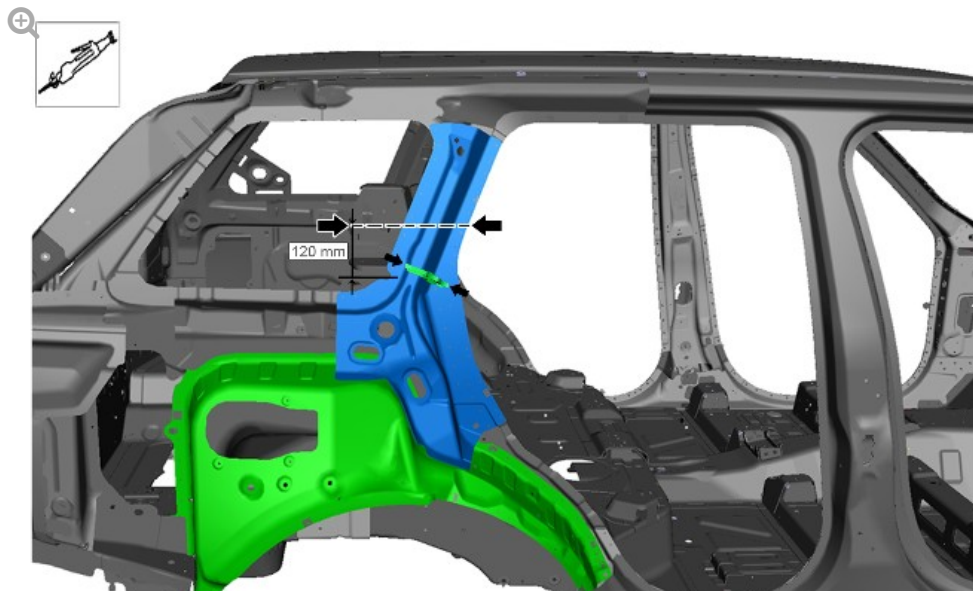
E143601

Remove the SPR's as indicated.

9.

CAUTION:

Care should be taken not to cut through into the inner panel.

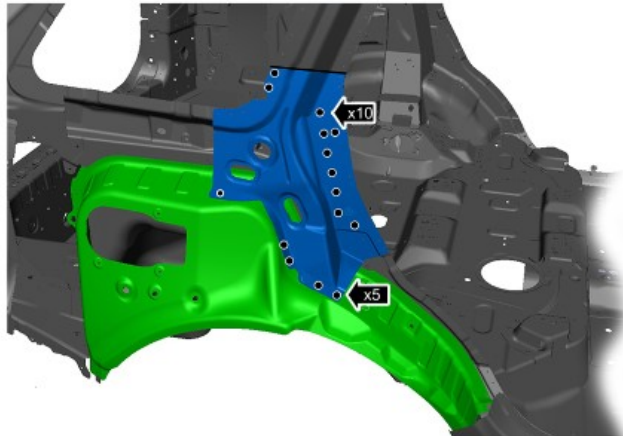


E143602

Release the noise, vibration and harshness (NVH) component and mark out the position where the C-pillar reinforcement metal inert gas (MIG) butt joint is to be made. Mark and cut through the C-

pillar reinforcement as indicated.

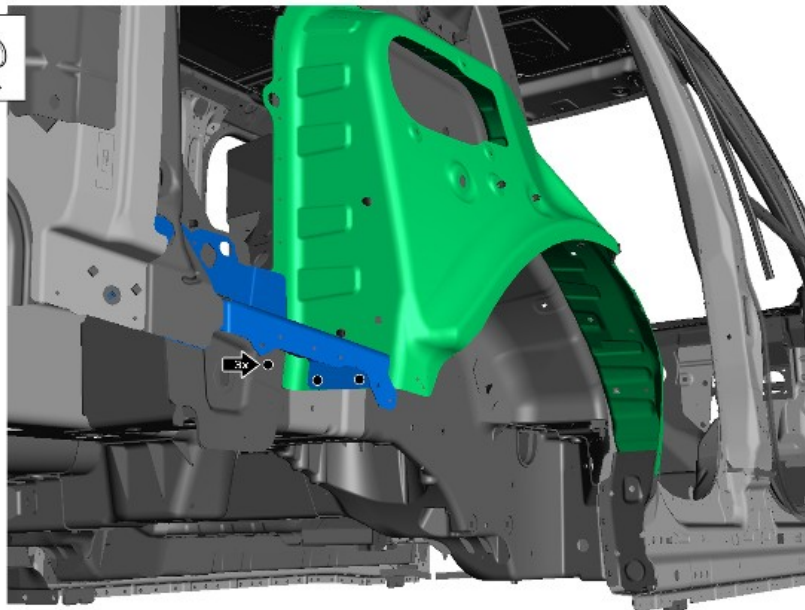
10.



E143603

Remove the SPR's as indicated.

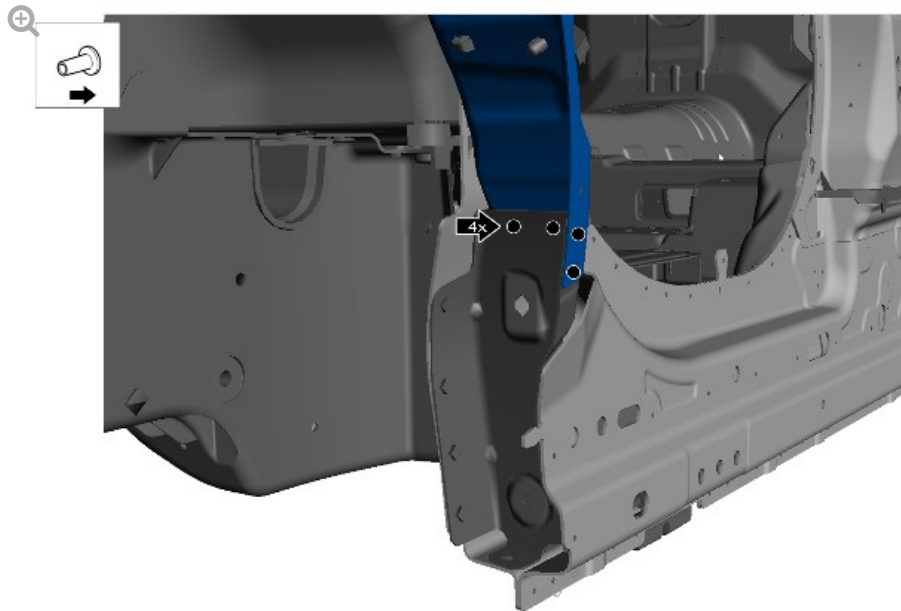
11.



E144566

Remove the SPR's as indicated.

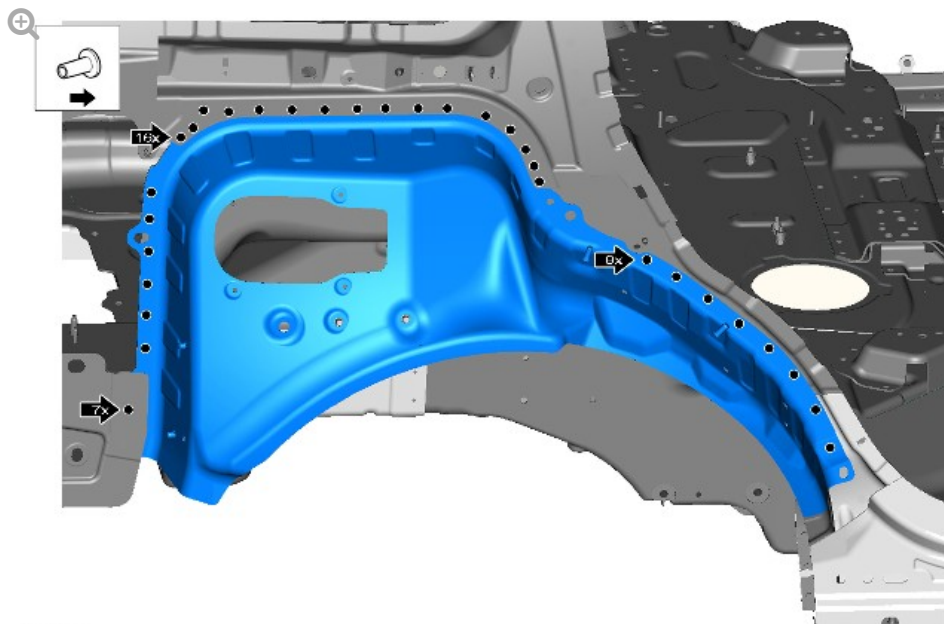
12.



E143605

Remove the SPR's as indicated.

13.

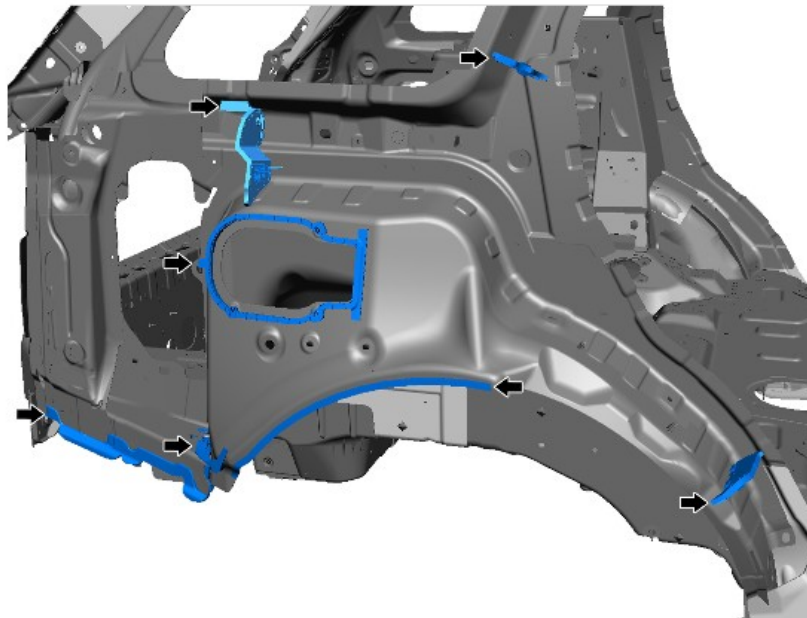


E143604

Remove the SPR's as indicated.

14. Separate the joints and remove the rear wheelhouse outer.

15.



E144564

Release the NVH components as indicated.

INSTALLATION

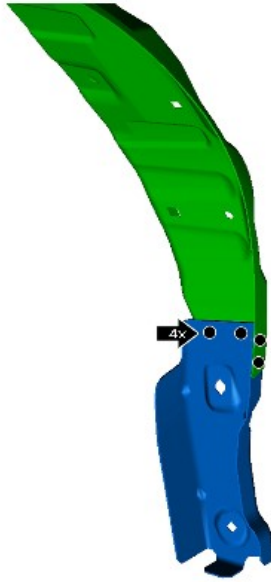
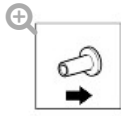
CAUTION:

Where structural adhesive is to be applied between surfaces it is essential compression is applied to the joints during the curing period.

NOTES:

- BSF installation involves using the Land Rover approved BSF installation tool and requires a hole diameter of 6.5mm.
 - SPR installation requires using the Land Rover approved SPR tool.
-

1.

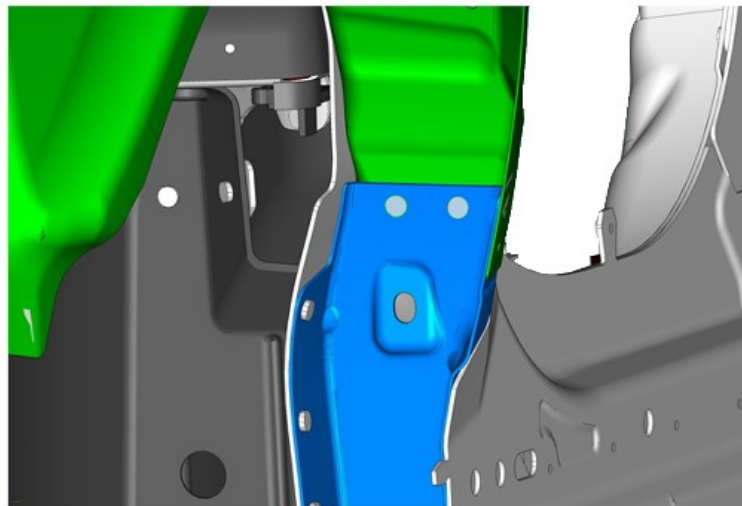


E143489

Remove the SPR's from the new rear wheelhouse front extension as indicated.

-
2. Remove rivet remnants.
 3. Dress flanges where necessary.
 4. Using a fine bristle disc, clean and prepare the panel surfaces.
 5. Carefully build an expansion space between the quarter panel inner and rear wheelhouse outer.
-

6.

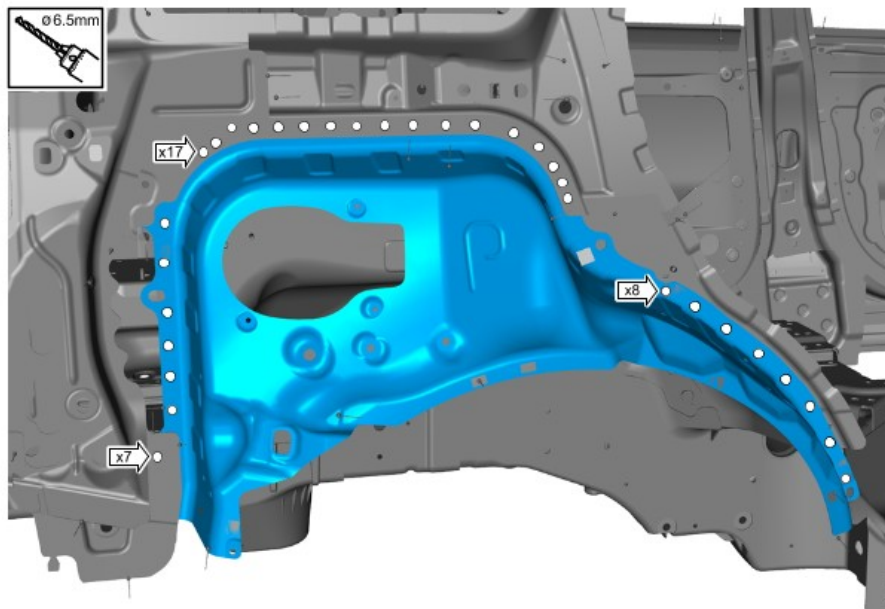


E144468

Make sure that the rear wheelhouse outer holes and the rear wheelhouse front extension holes line up correctly as indicated.

-
7. Offer up the new rear wheelhouse outer and clamp into position. Check alignment, if correct proceed to next step, if not, rectify and recheck before proceeding.
-

8.



E144559

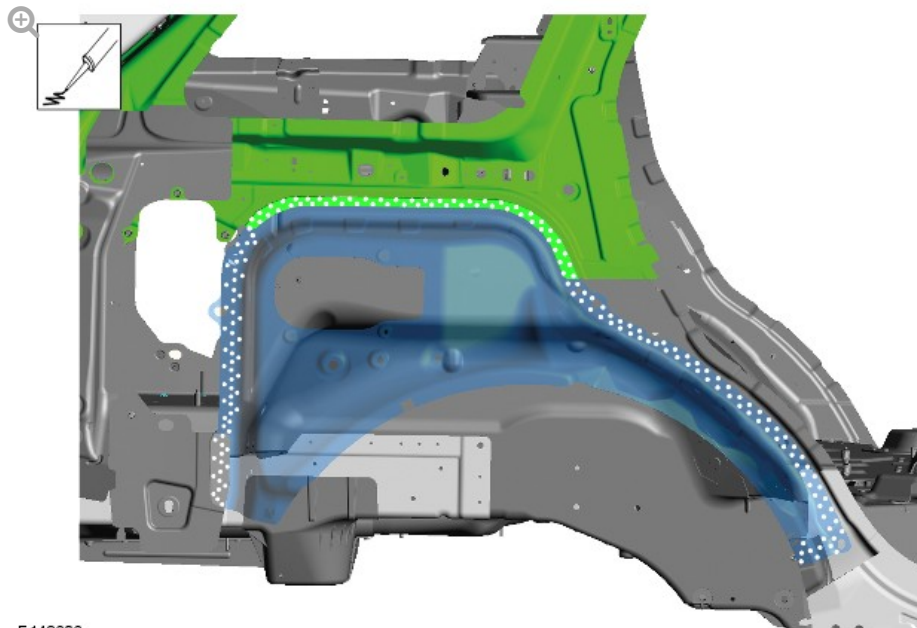
Drill through the old rear wheelhouse outer panel holes into the new rear wheelhouse outer panel where BSF's are to be installed as indicated.

9. Remove the new rear wheelhouse outer.
10. Deburr the drilled holes.
11. Using a fine bristle disc, clean and prepare the panel surfaces.
12. Apply the coupling agent where the Land Rover approved bonding material is to be applied and allow to dry.

13.

NOTE:

Make sure a continuous bead of adhesive surrounds fixing holes.



E143606

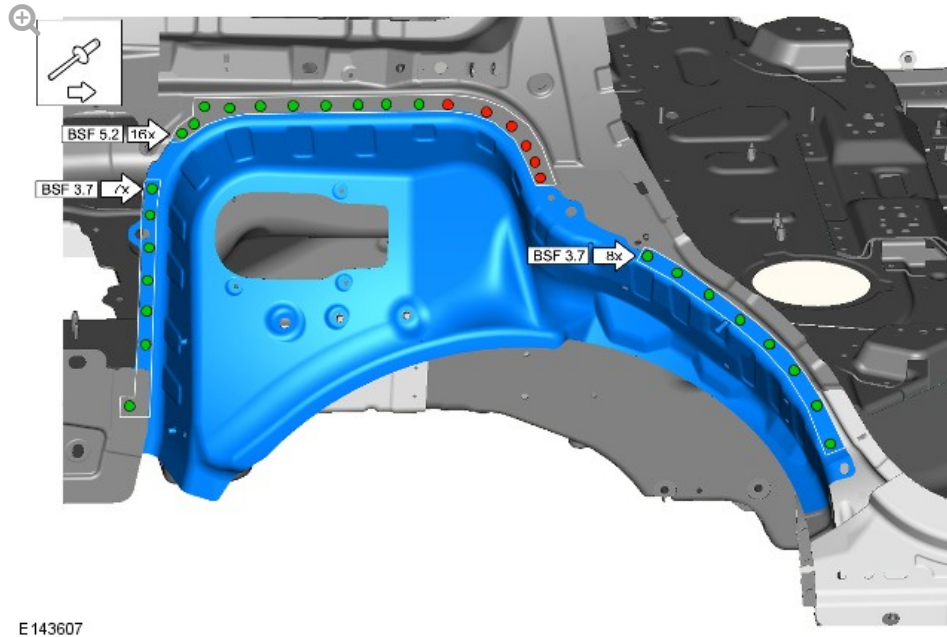
Apply a 5mm bead of Land Rover recommended bonding material as indicated.

14. Offer up the new rear wheelhouse outer, clamp into position. Check alignment, if correct, proceed to next step, if not rectify and recheck before proceeding.

15.

NOTES:

- Make sure that the BSF's indicated in **green** are installed from the **inside**.
- Make sure that the BSF's indicated in **red** are installed from the **outside**.

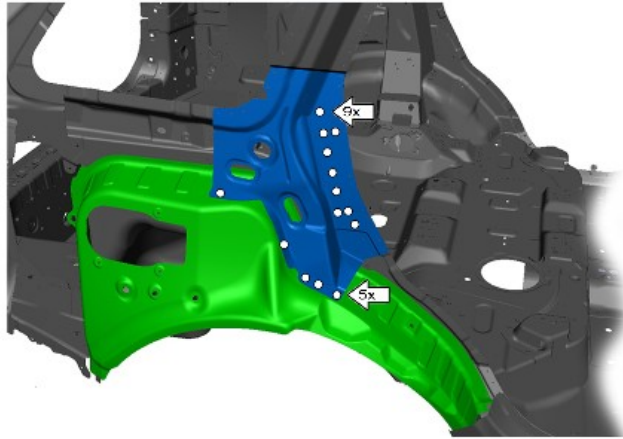


Install the BSF's as indicated.

16. Remove any excess adhesive.

17. Offer up the C-pillar reinforcement section, align and clamp into position. Check alignment, if correct, proceed to next step, if not rectify and recheck before proceeding.

18.



E144560

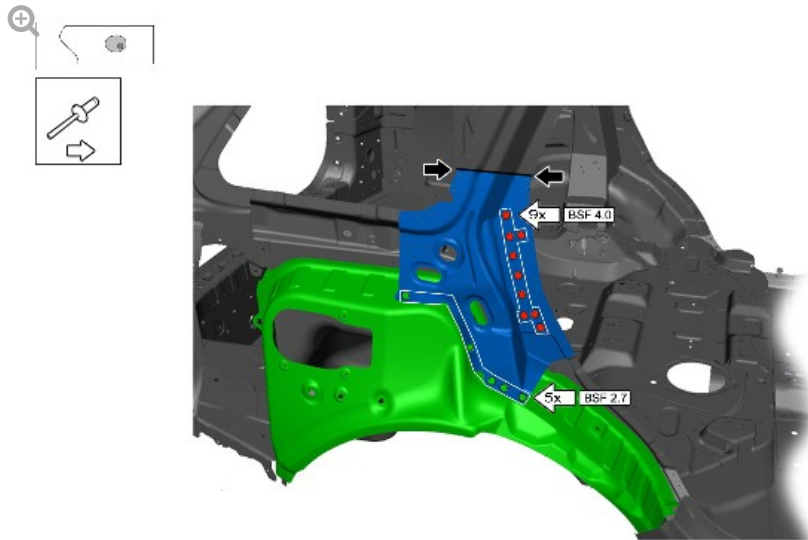
Drill through the old C-pillar reinforcement where BSF's are to be installed as indicated.

19. Deburr the drilled holes.

20.

NOTES:

- Make sure that the BSF's indicated in **green** are installed from the **inside**.
- Make sure that the BSF's indicated in **red** are installed from the **outside**.

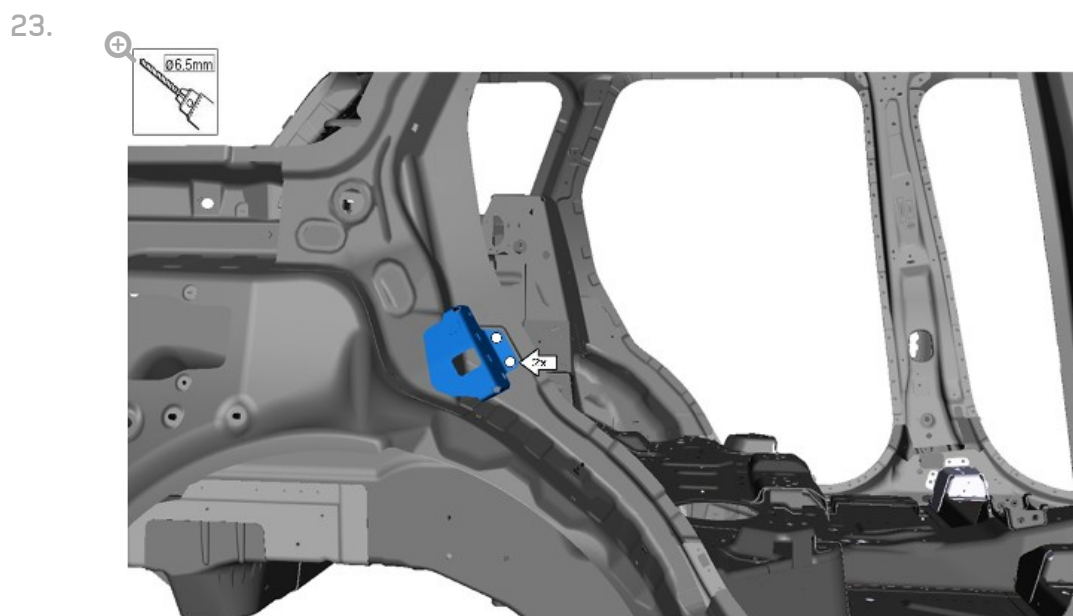


E 143608

MIG weld the butt joint and install the BSF's as indicated.

21. Carry out a non destructive crack inspection on the MIG butt joint.
If correct proceed to next step, if not, rectify and recheck before proceeding.

22. Dress the welded joint.



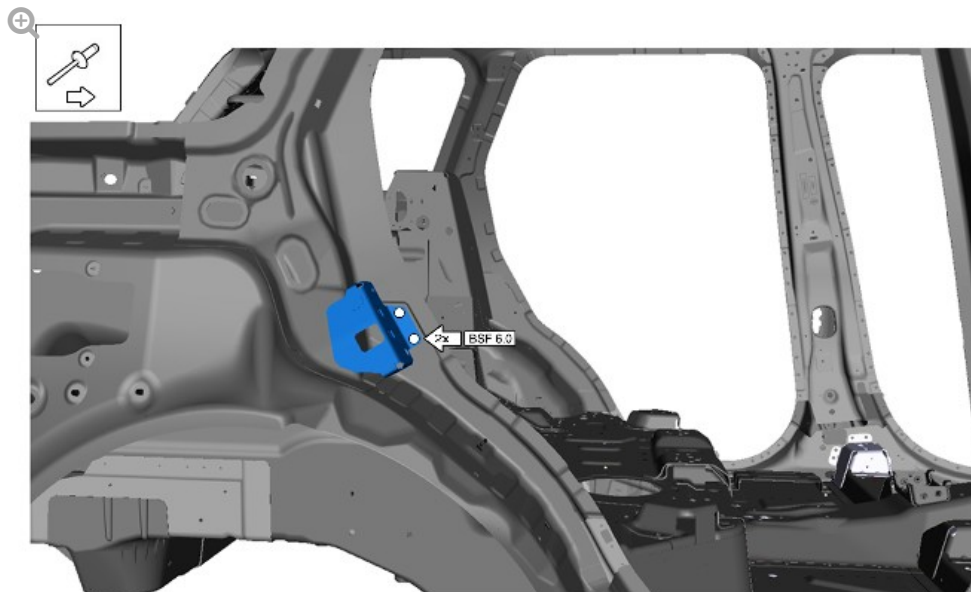
E 144561

Drill holes through the old door striker plate mounting plate into

the new rear wheelhouse outer where BSF's are to be installed.

24. Deburr the drilled holes.

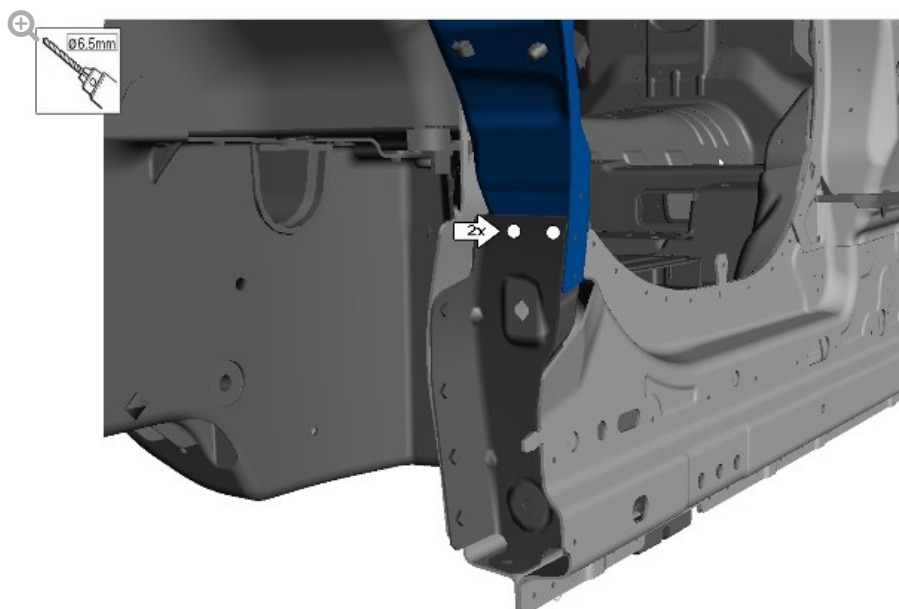
25.



E144558

Install the BSF's as indicated.

26.

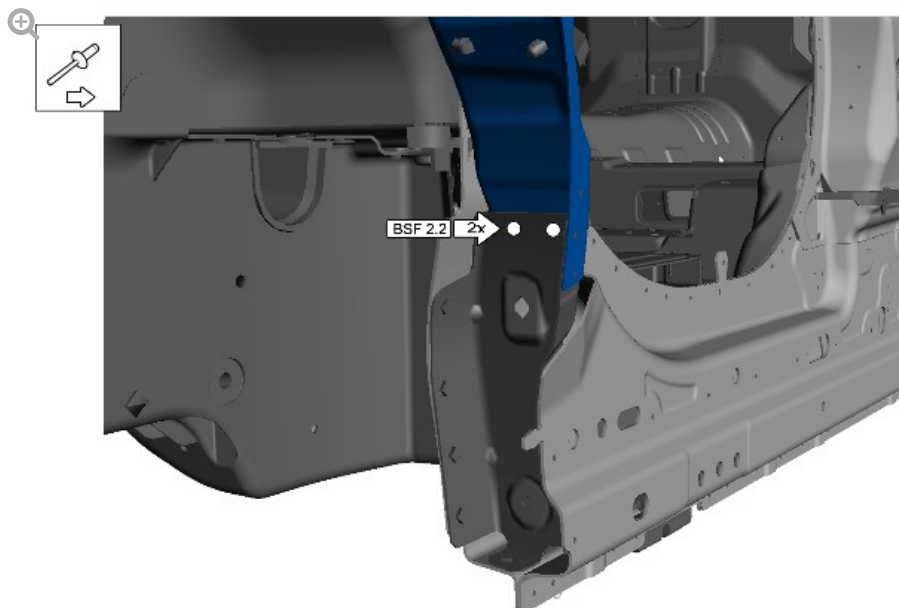


E144562

Drill holes through the old rear wheelhouse outer into the new rear wheelhouse outer panel where BSF's are to be installed as indicated.

27. Deburr the drilled holes.

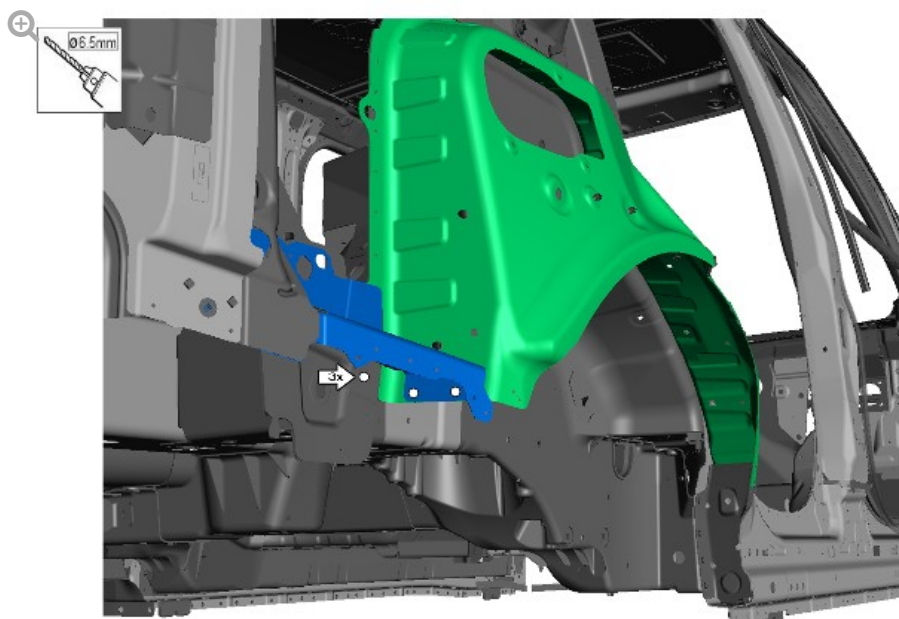
28.



E 144563

Install the BSF's as indicated.

29.

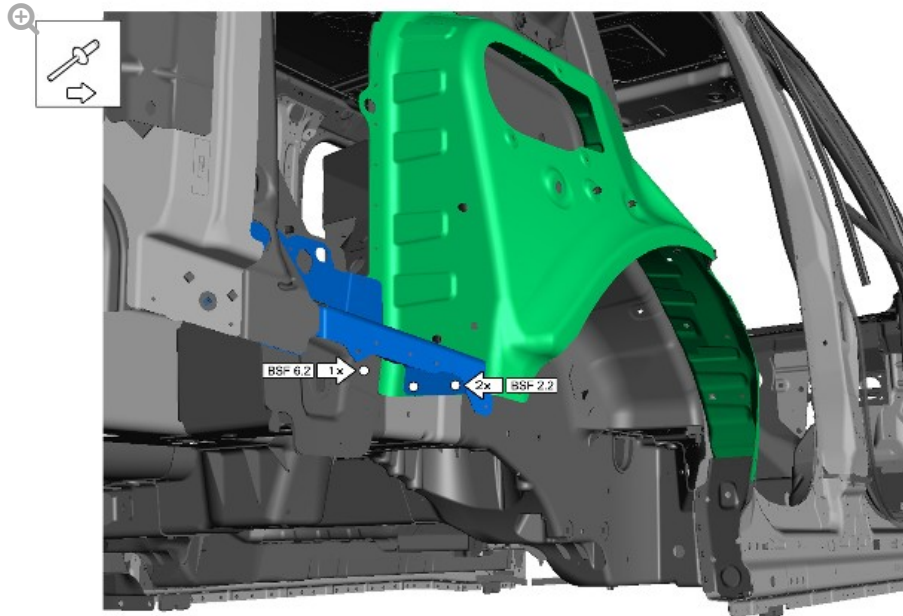


E 144565

Drill holes through the rear wheelhouse extension into the rear wheelhouse outer where BSF's are to be installed as indicated.

30. Deburr the drilled holes.

31.

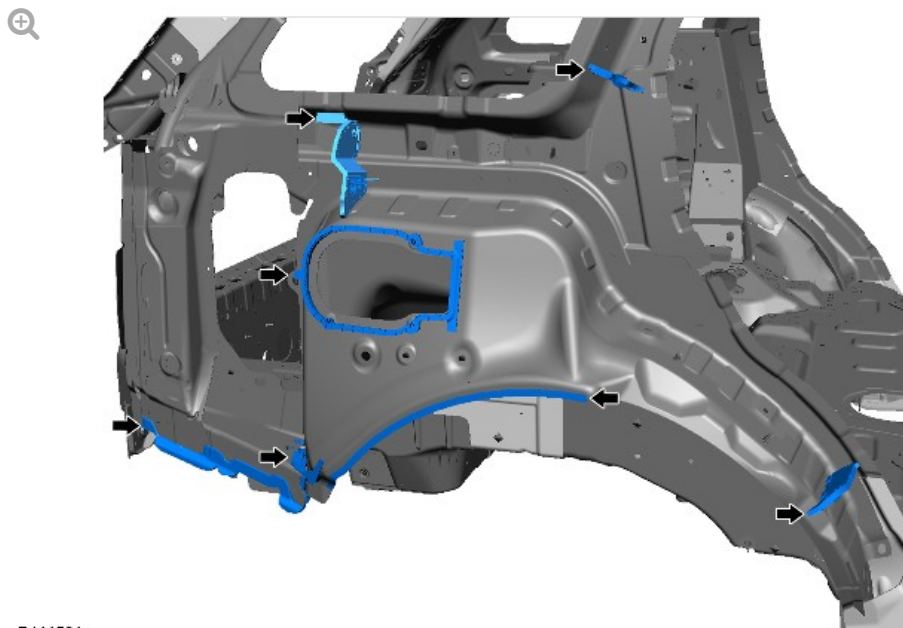


E 144567

Install the BSF's as indicated.

32. Remove any excess adhesive.

33.



E 144564

Install the NVH components as indicated.

34. Make sure that any open or exposed panel joints are correctly sealed.

35. Make sure corrosion protection is applied to all areas affected by repair.

36. The installation of associated panels and components is the reversal of removal procedure.